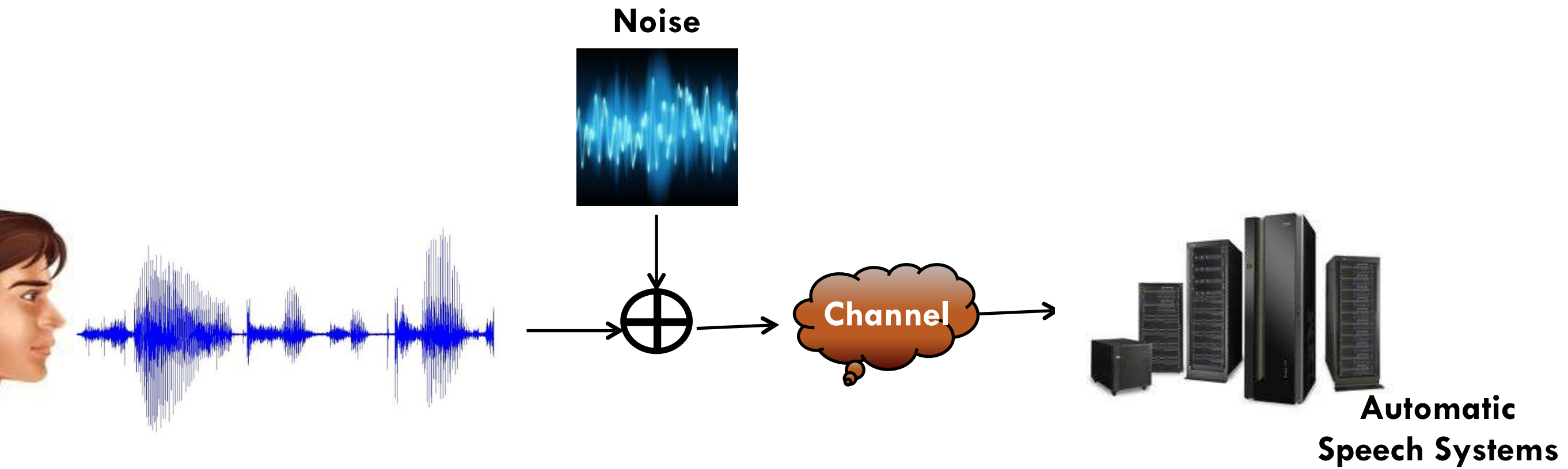


E9 205 Machine Learning for Signal Processing

Deep Learning for Audio and Vision

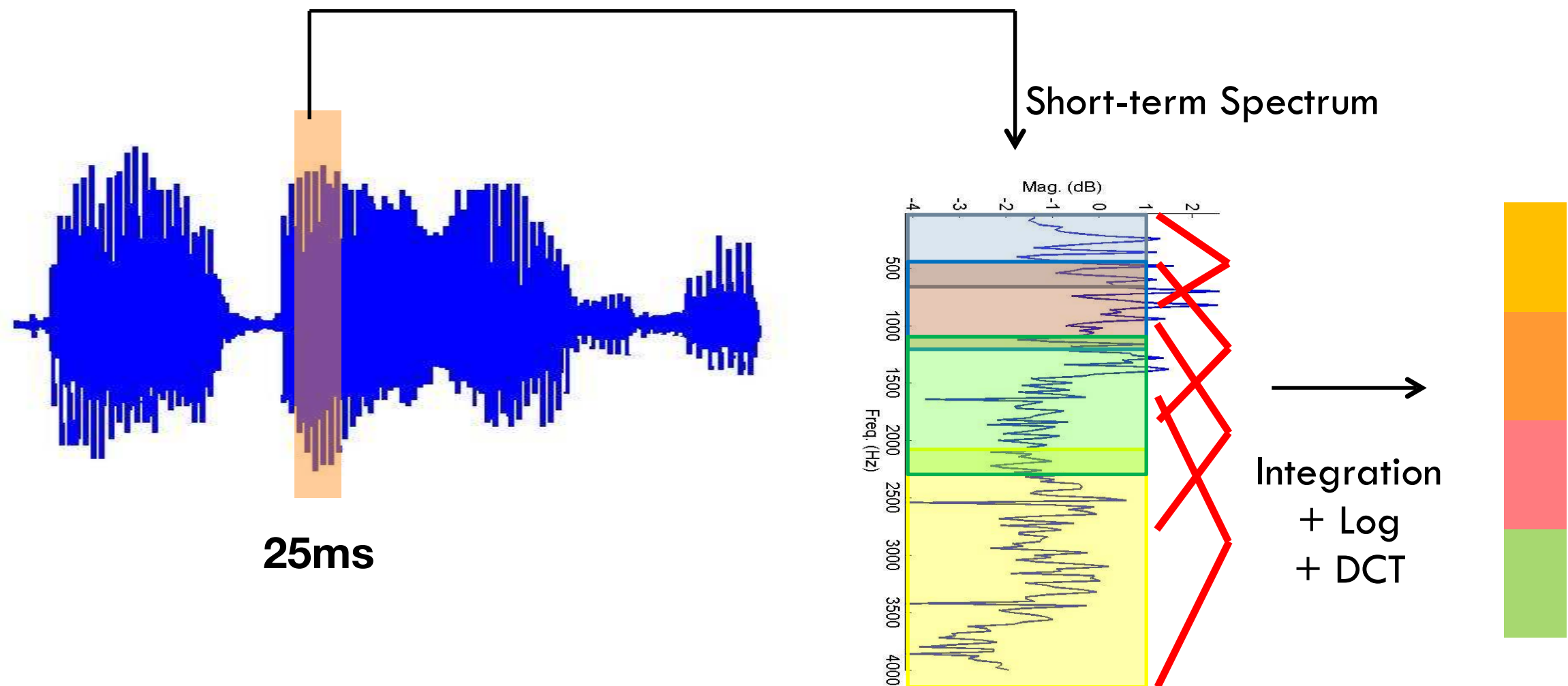
20-11-2019

Speech Recognition



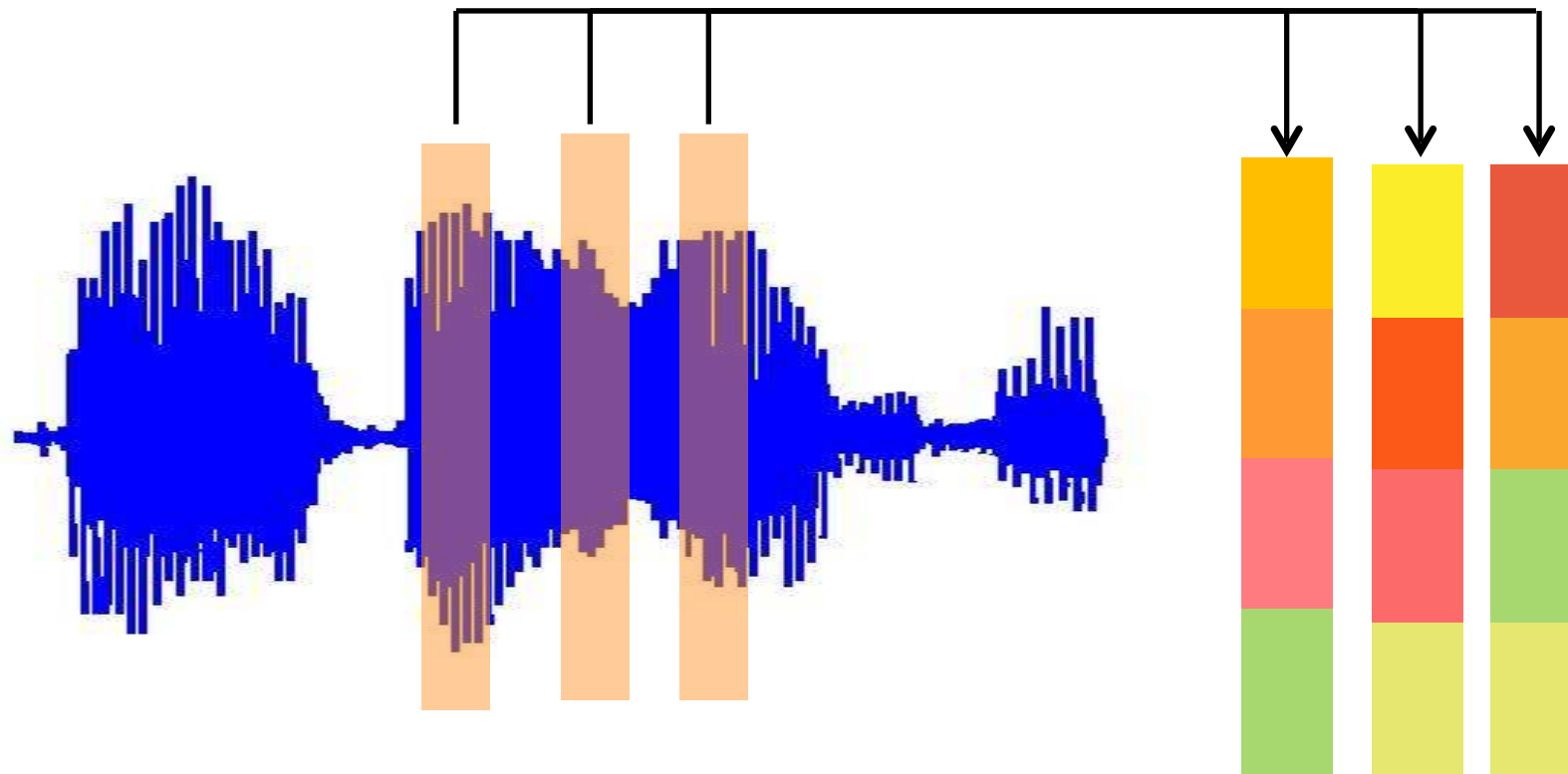
Signal Modeling

- Short-term spectra integrated in mel frequency bands followed by log compression + DCT – **mel frequency cepstral coefficients (MFCC)** [Davis and Mermelstein, 1979].



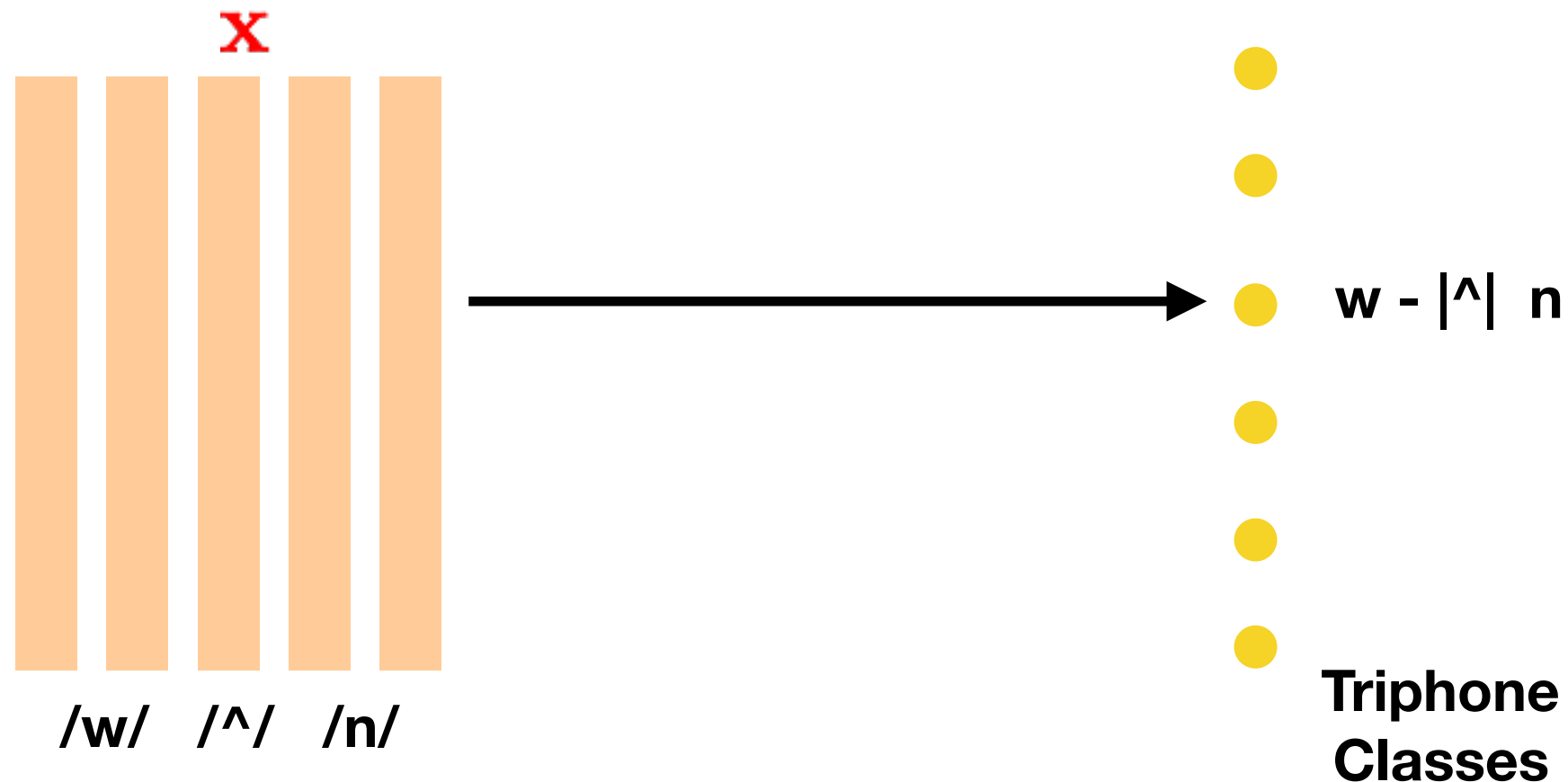
Mel Frequency Cepstral Coefficients

- MFCC processing repeated for every short-term frame yielding a sequence of features. Typically 25ms frames with 10ms hop in time.



Speech Recognition

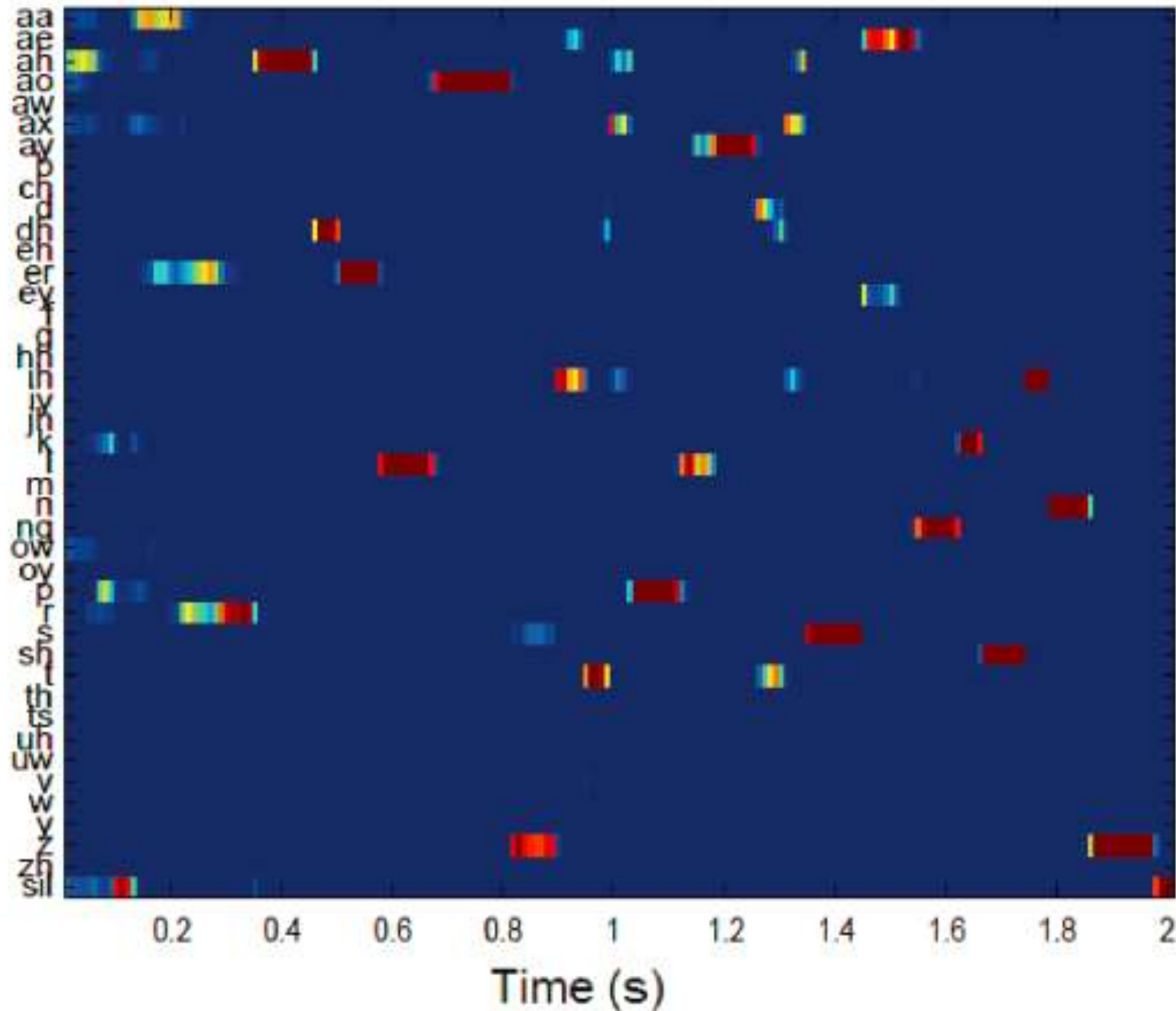
- Map the features to phone class. Using phone labelled data.



- Classical machine learning - train a classifier on speech training data that maps to the target phoneme class.

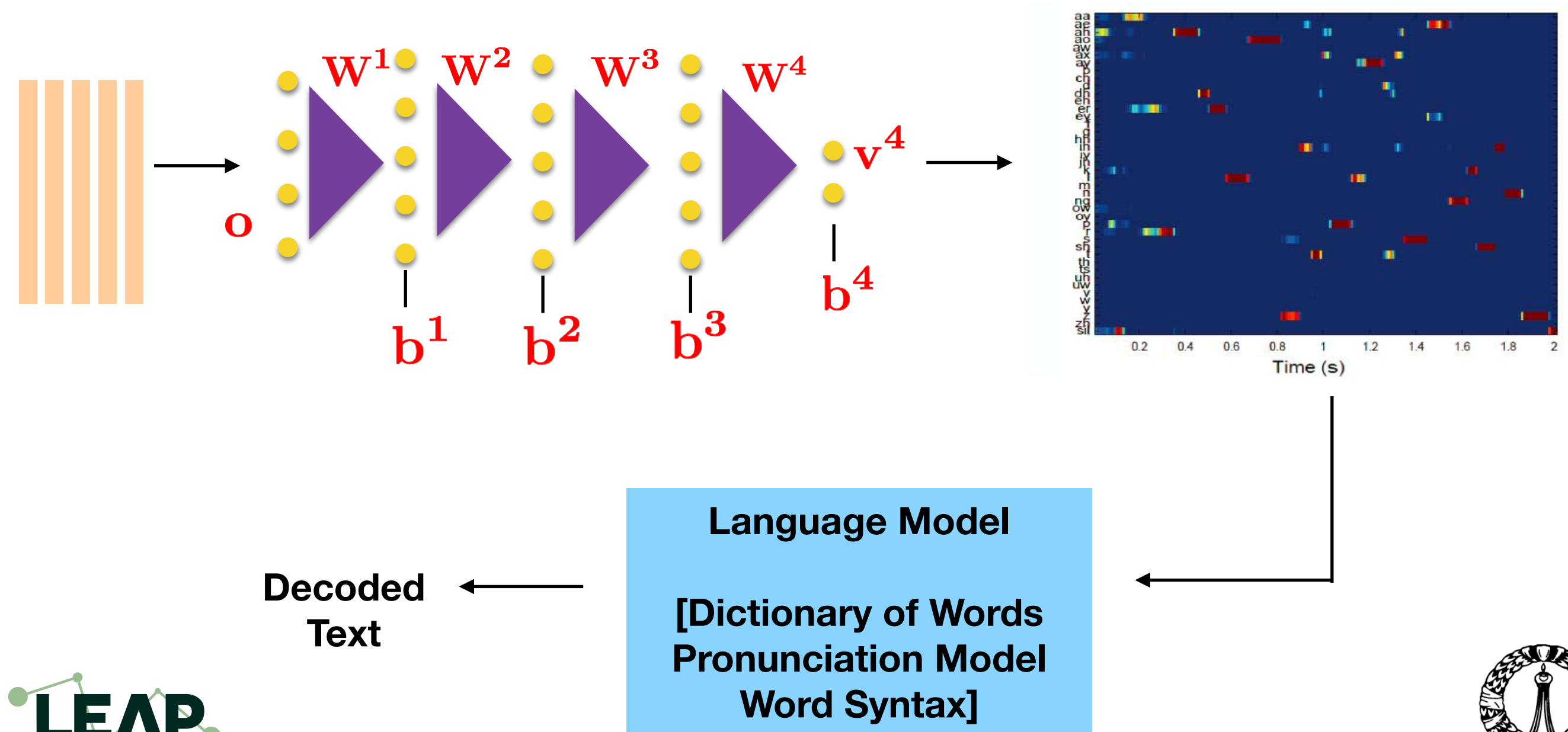
Back to Speech Recognition

Ma

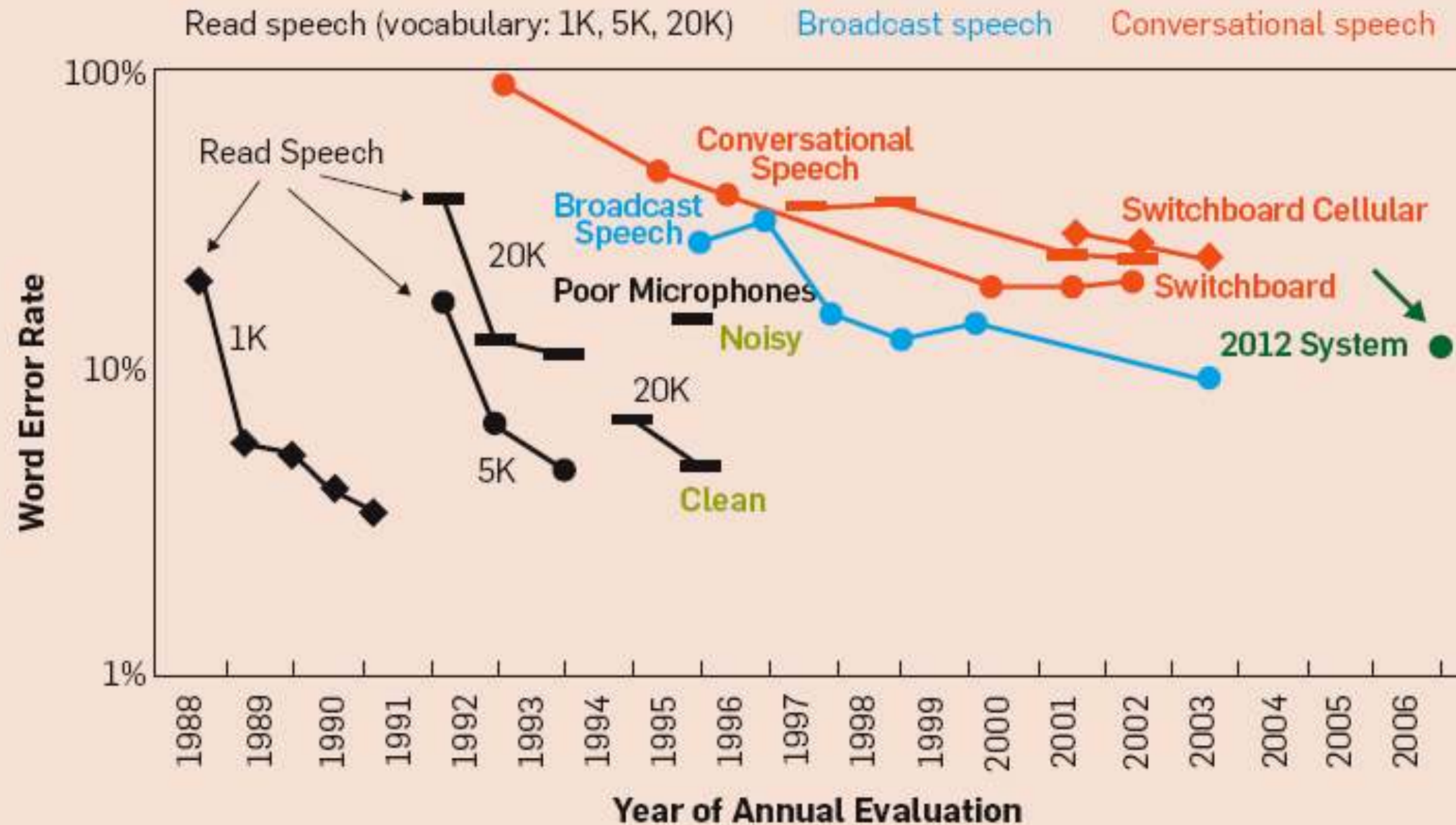


Back to Speech Recognition

Mapping Speech Features to Phonemes to words



State of Progress



2018

5.3%

Claims of human parity using
BLSTM based Models !!!

Moving to End-to-End

Text Output

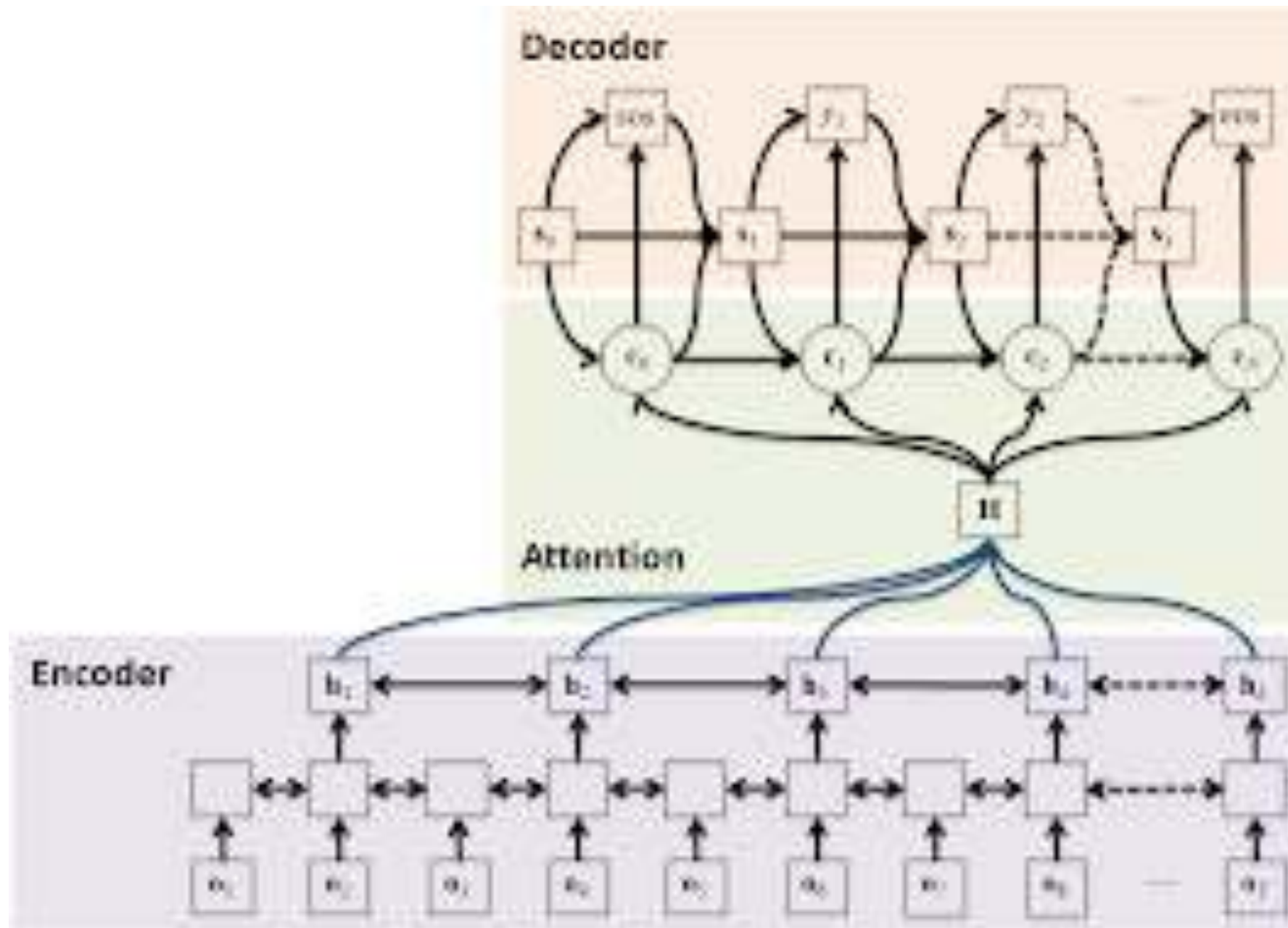
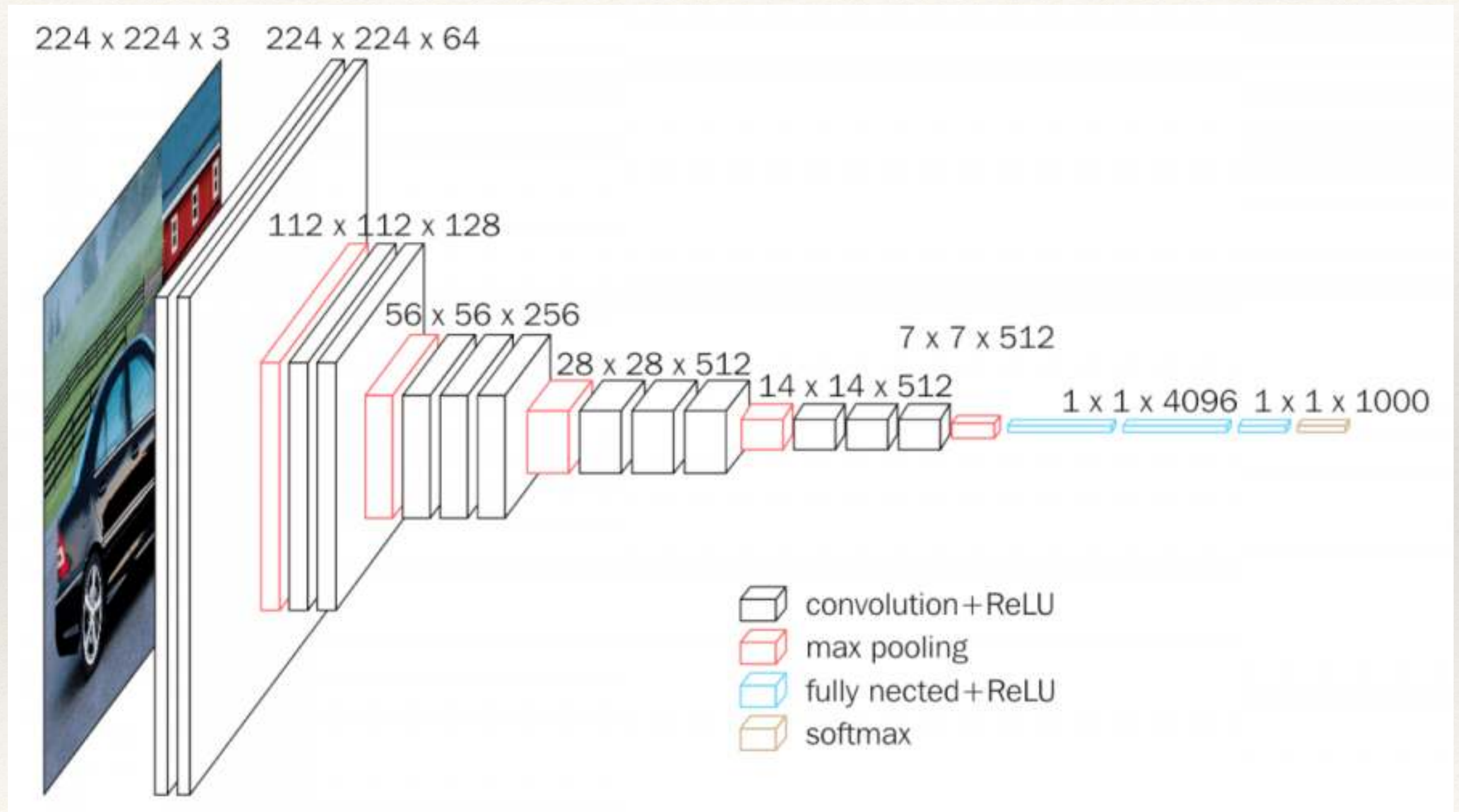


Image Processing

Visual Graphics Group Network



ImageNet Task

1000 images in each of 1000 categories. In all, there are roughly 1.2 million training images, 50,000 validation images, and 150,000 testing images. ImageNet consists of variable-resolution images. Therefore, the images have been down-sampled to a fixed resolution of 224×224 .

Method	top-1 val. error (%)	top-5 val. error (%)	top-5 test error (%)
VGG (2 nets, multi-crop & dense eval.)	23.7	6.8	6.8
VGG (1 net, multi-crop & dense eval.)	24.4	7.1	7.0
VGG (ILSVRC submission, 7 nets, dense eval.)	24.7	7.5	7.3
GoogLeNet (Szegedy et al., 2014) (1 net)	-	7.9	
GoogLeNet (Szegedy et al., 2014) (7 nets)	-	6.7	
MSRA (He et al., 2014) (11 nets)	-	-	8.1
MSRA (He et al., 2014) (1 net)	27.9	9.1	9.1
Clarifai (Russakovsky et al., 2014) (multiple nets)	-	-	11.7
Clarifai (Russakovsky et al., 2014) (1 net)	-	-	12.5
Zeiler & Fergus (Zeiler & Fergus, 2013) (6 nets)	36.0	14.7	14.8
Zeiler & Fergus (Zeiler & Fergus, 2013) (1 net)	37.5	16.0	16.1
OverFeat (Sermanet et al., 2014) (7 nets)	34.0	13.2	13.6
OverFeat (Sermanet et al., 2014) (1 net)	35.7	14.2	-
Krizhevsky et al. (Krizhevsky et al., 2012) (5 nets)	38.1	16.4	16.4
Krizhevsky et al. (Krizhevsky et al., 2012) (1 net)	40.7	18.2	-

Can we go deeper

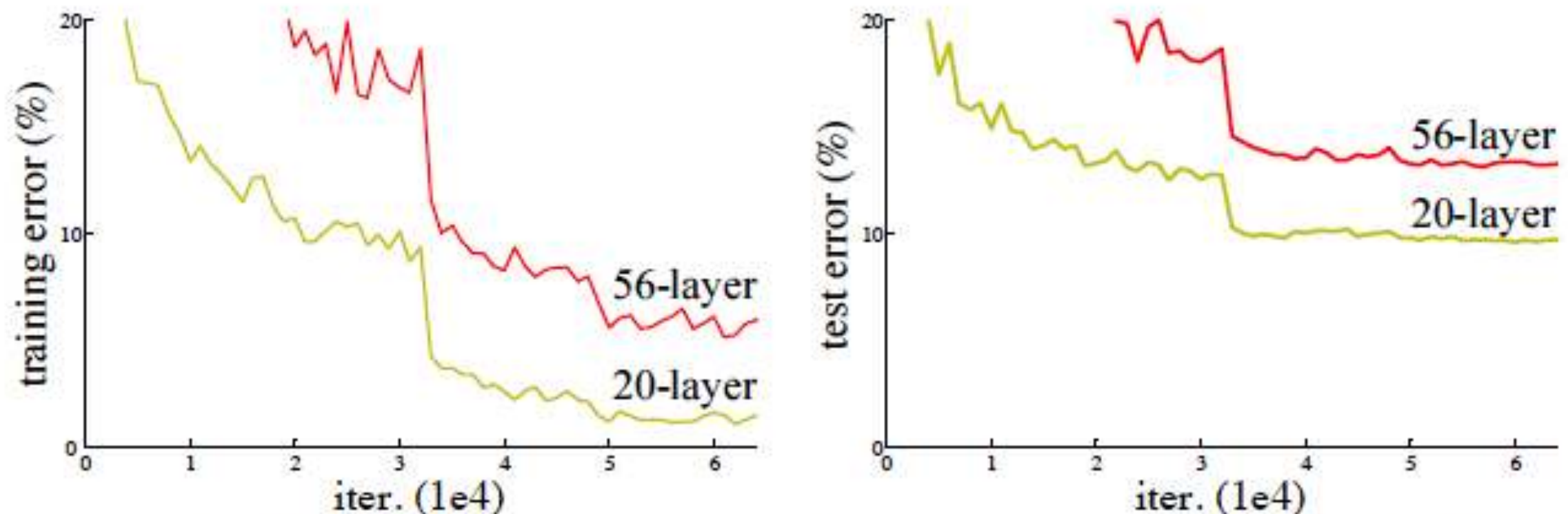


Figure 1. Training error (left) and test error (right) on CIFAR-10 with 20-layer and 56-layer “plain” networks. The deeper network has higher training error, and thus test error. Similar phenomena on ImageNet is presented in Fig. 4.

Deep Residual Learning for Image Recognition

Kaiming He Xiangyu Zhang Shaoqing Ren Jian Sun

Microsoft Research

{kahe, v-xiangz, v-shren, jiansun}@microsoft.com

Residual Blocks

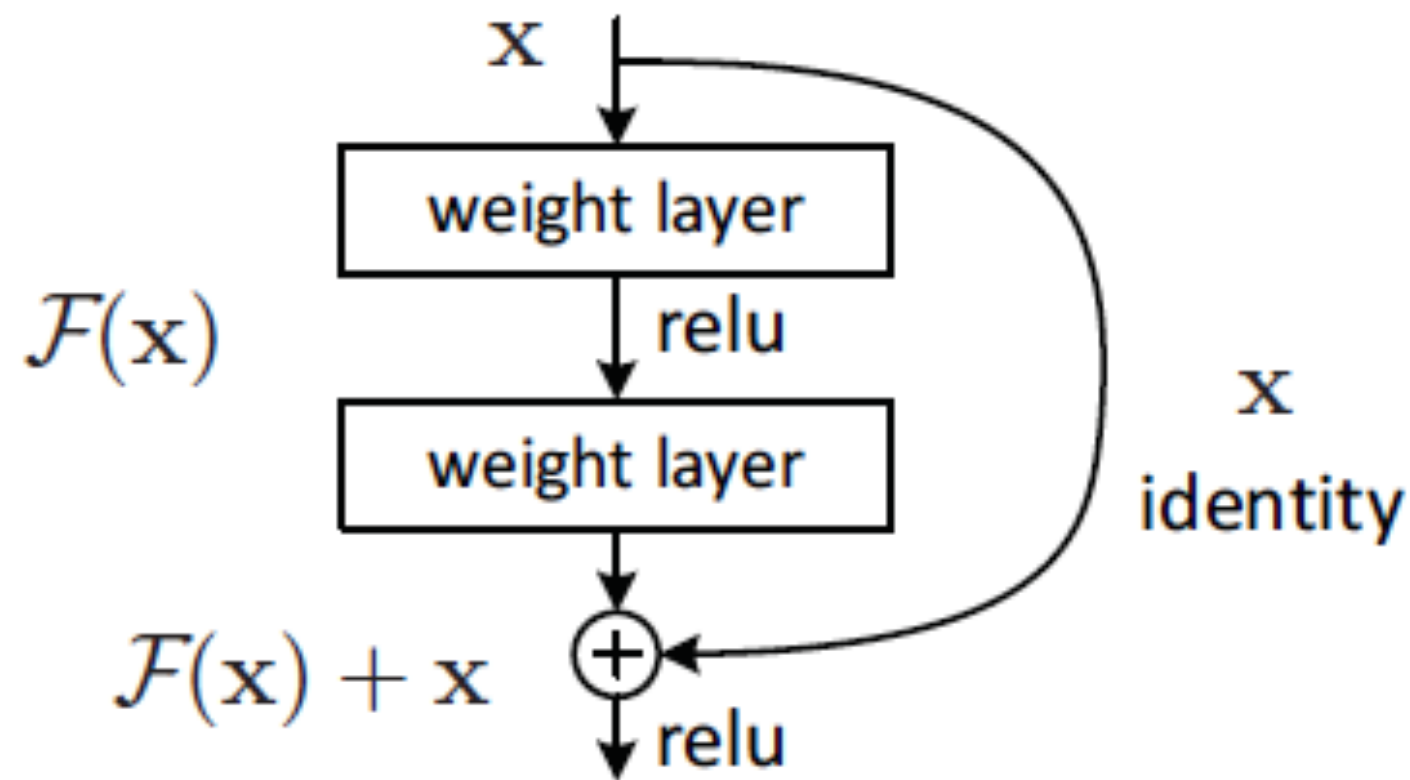
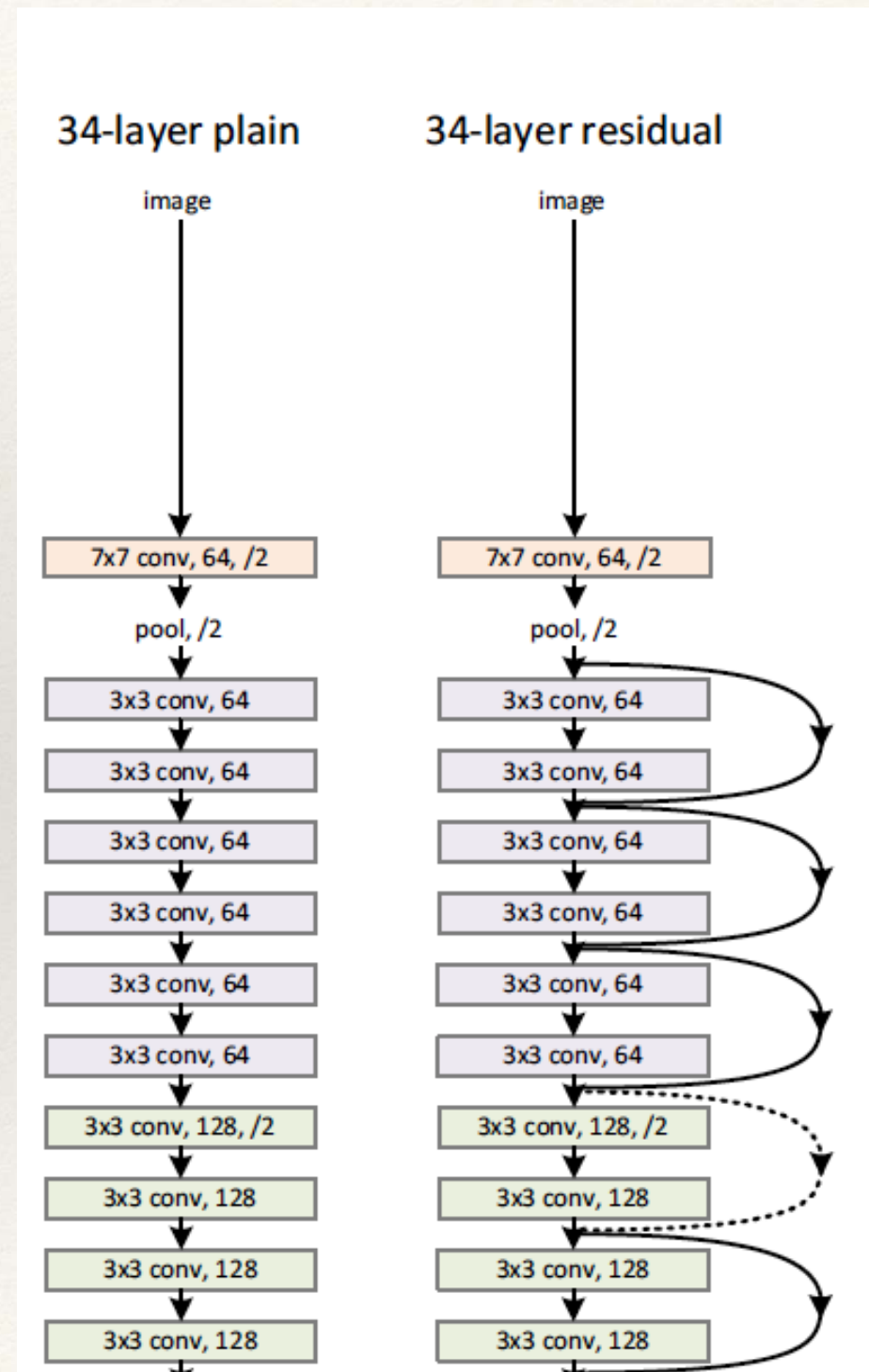


Figure 2. Residual learning: a building block.

Deep Networks with Residual Blocks



Deep Networks with Residual Blocks

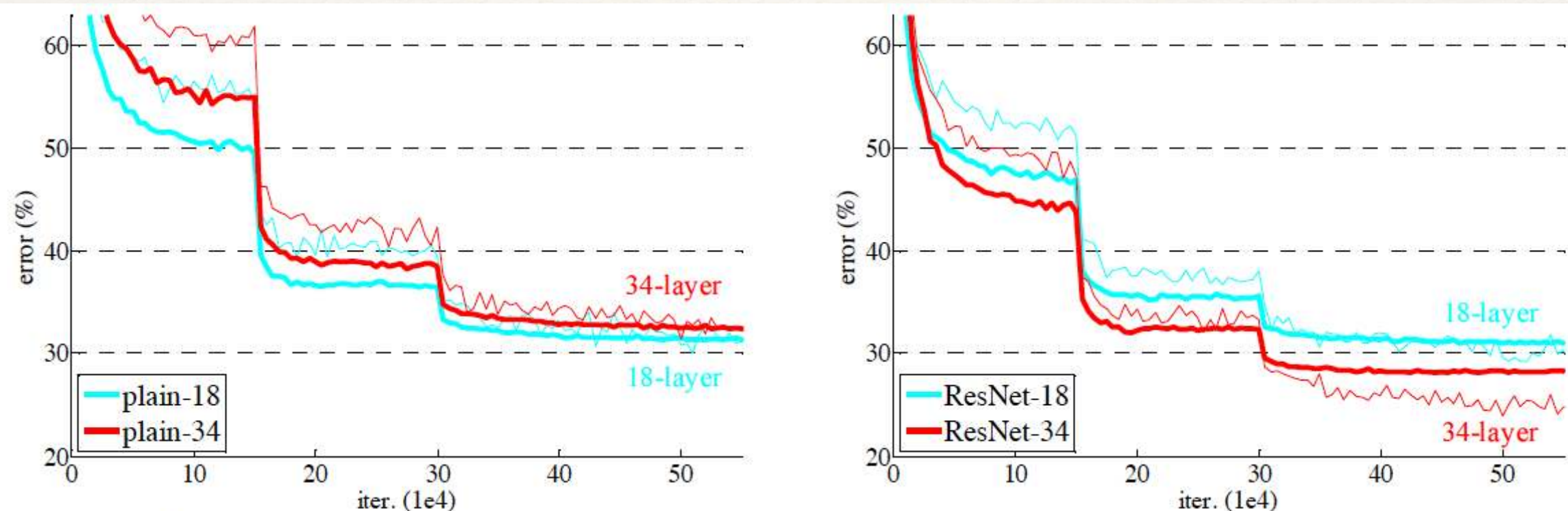


Figure 4. Training on **ImageNet**. Thin curves denote training error, and bold curves denote validation error of the center crops. Left: plain networks of 18 and 34 layers. Right: ResNets of 18 and 34 layers. In this plot, the residual networks have no extra parameter compared to their plain counterparts.

Results with ResNet

	plain	ResNet
18 layers	27.94	27.88
34 layers	28.54	25.03

Table 2. Top-1 error (% , 10-crop testing) on ImageNet validation. Here the ResNets have no extra parameter compared to their plain counterparts. Fig. 4 shows the training procedures.

ResNet-50	21.12	7.10
ResNet-50	22.85	6.71
ResNet-101	21.75	6.05
ResNet-152	21.43	5.71

Image Segmentation

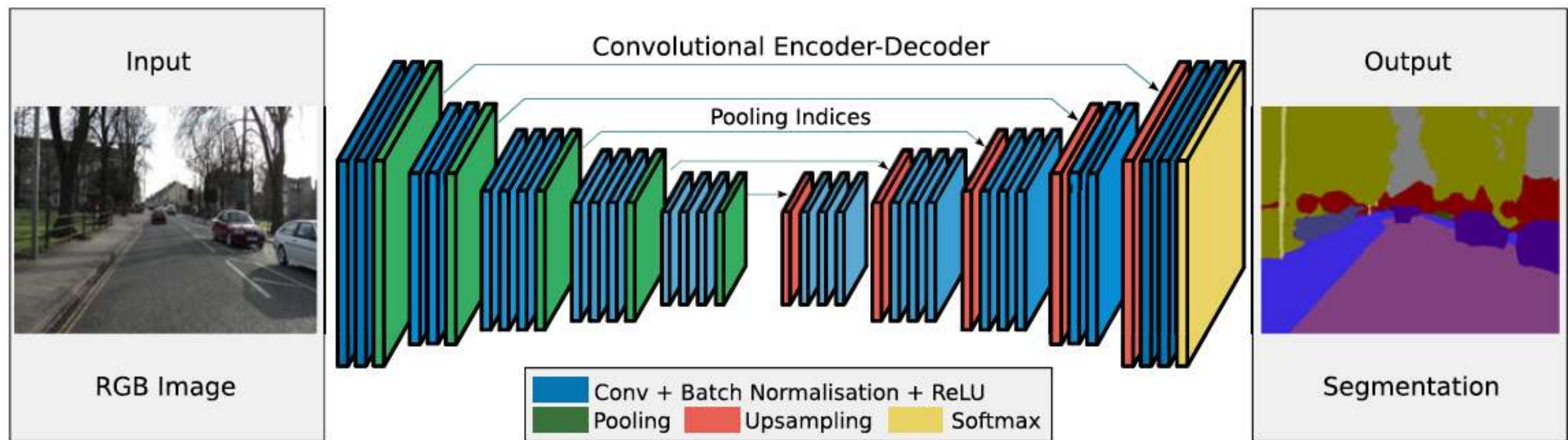
SegNet: A Deep Convolutional Encoder-Decoder Architecture for Image Segmentation

Vijay Badrinarayanan, Alex Kendall , and Roberto Cipolla, *Senior Member, IEEE*

The Problem of Segmentation



SegNet Architecture

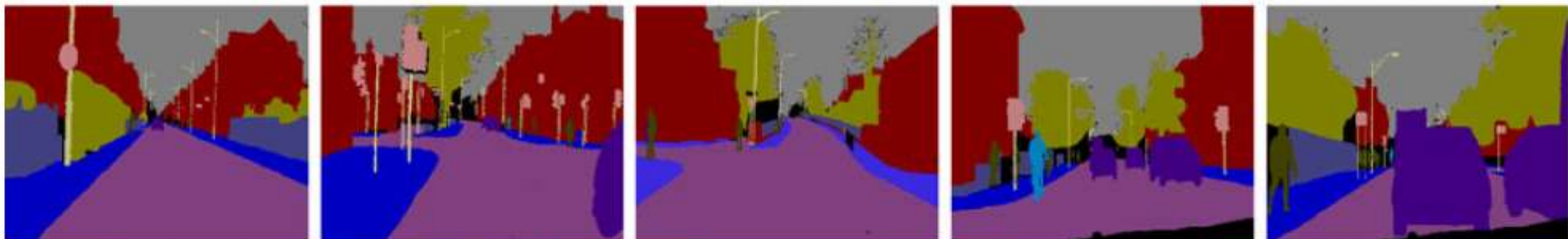


Results from Segnet

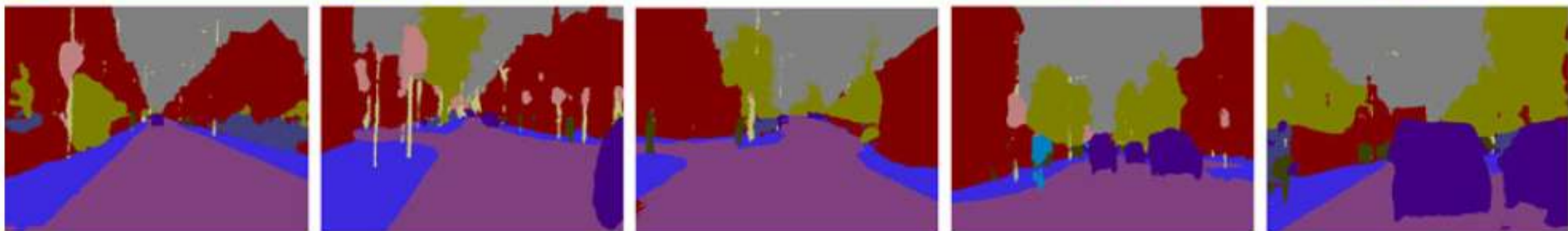
Test samples



Ground Truth

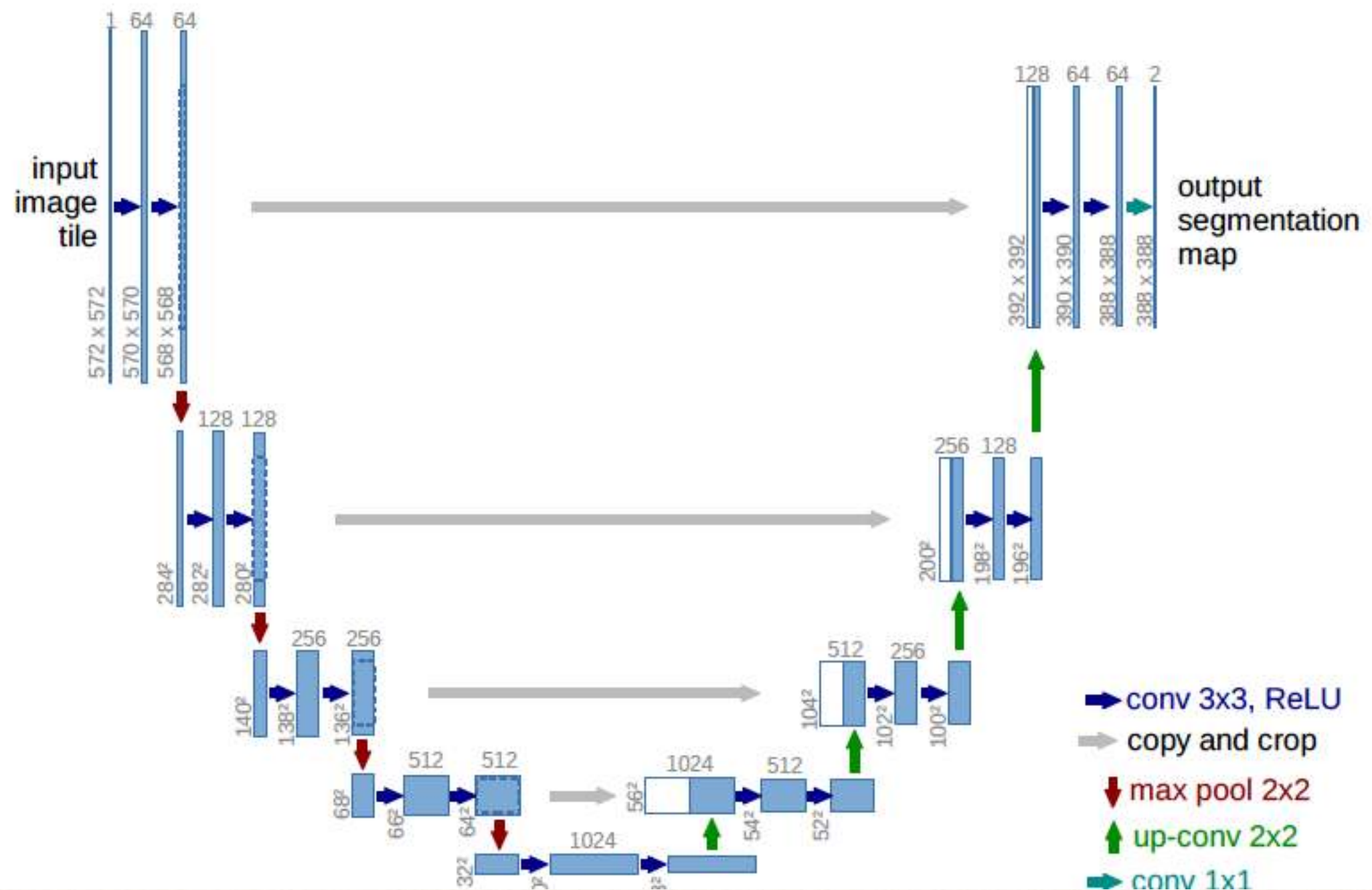


SegNet



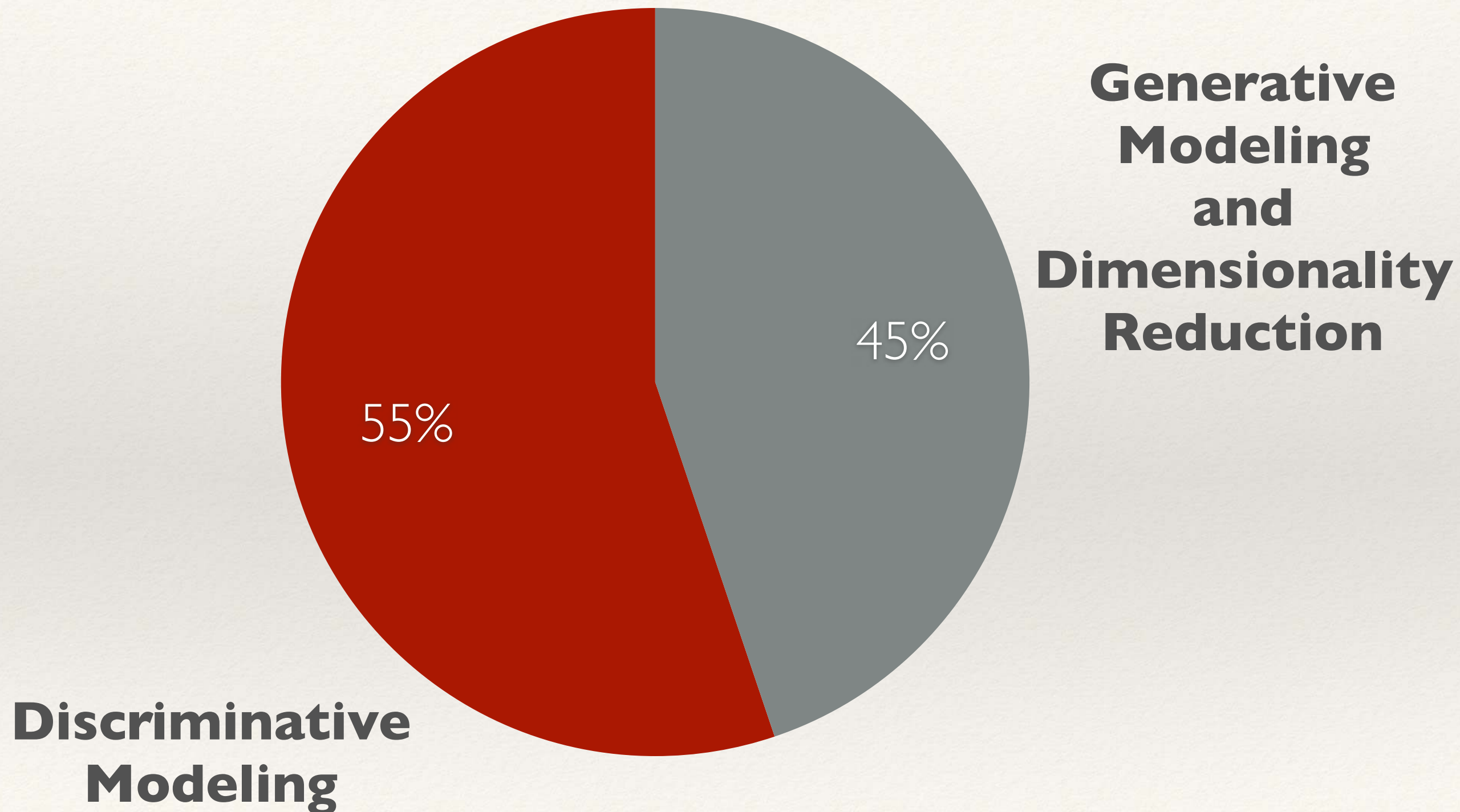
U-Net: Convolutional Networks for Biomedical Image Segmentation

U-net



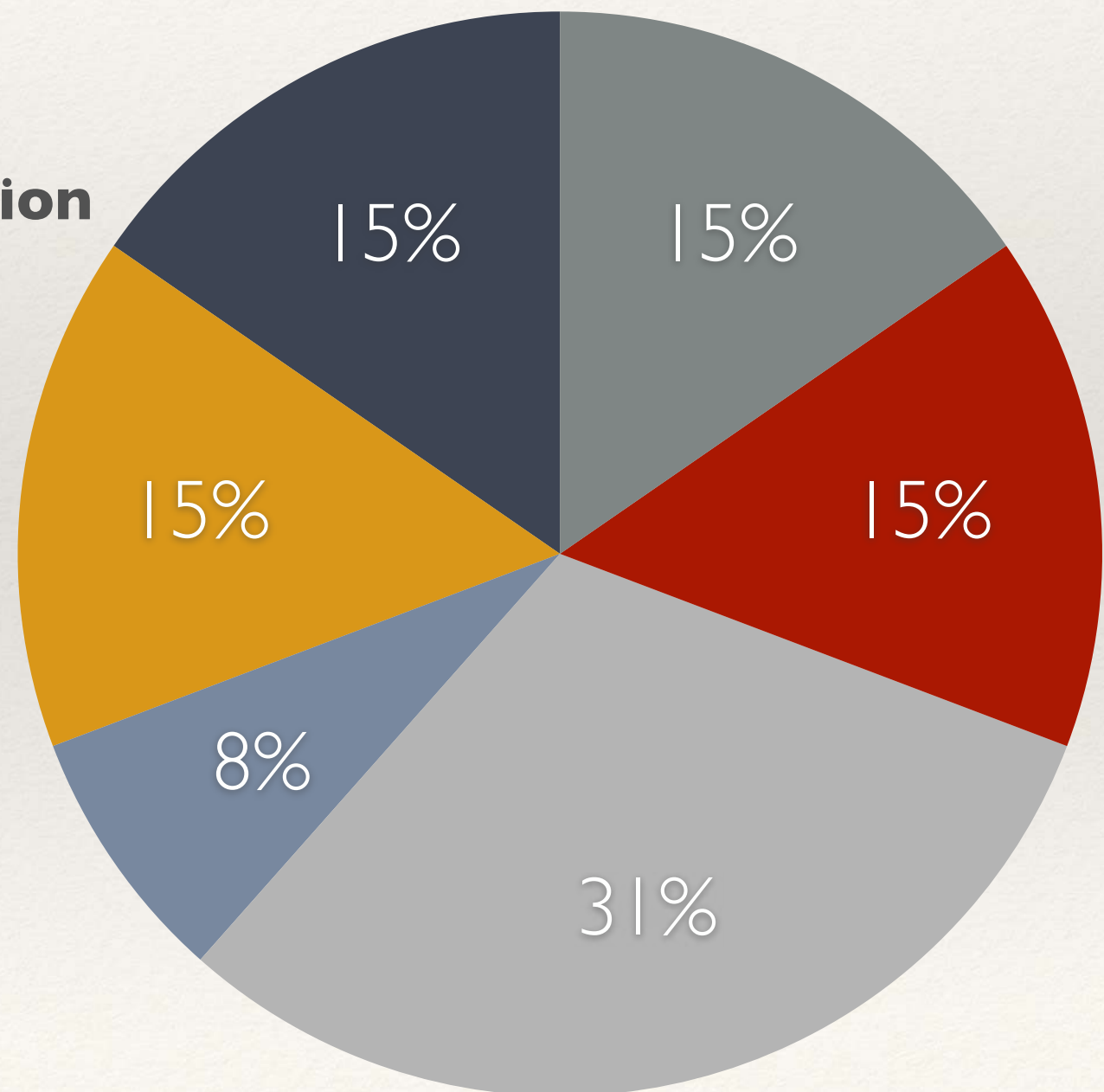
Summary of the Course

Distribution Pie Chart

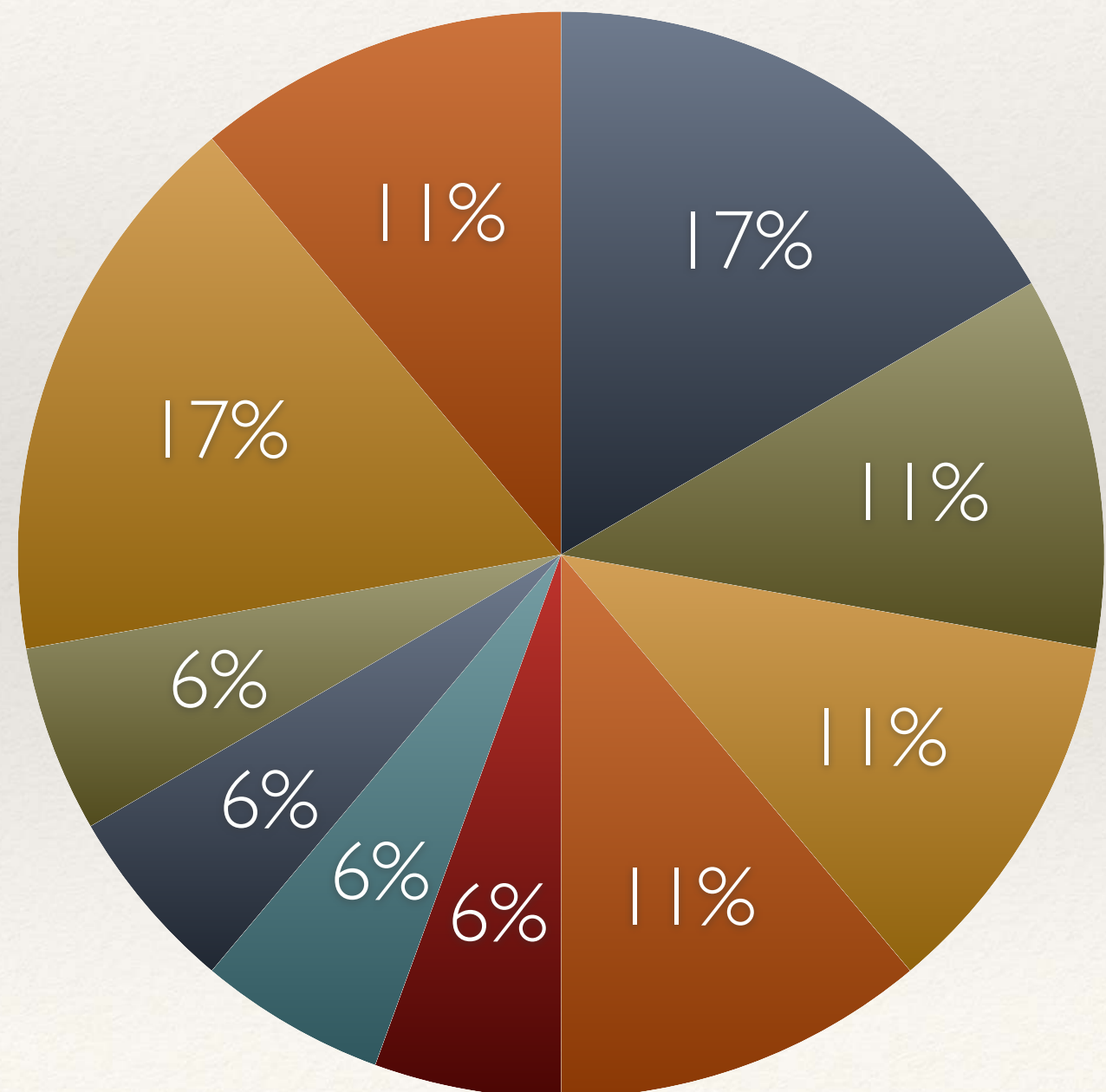
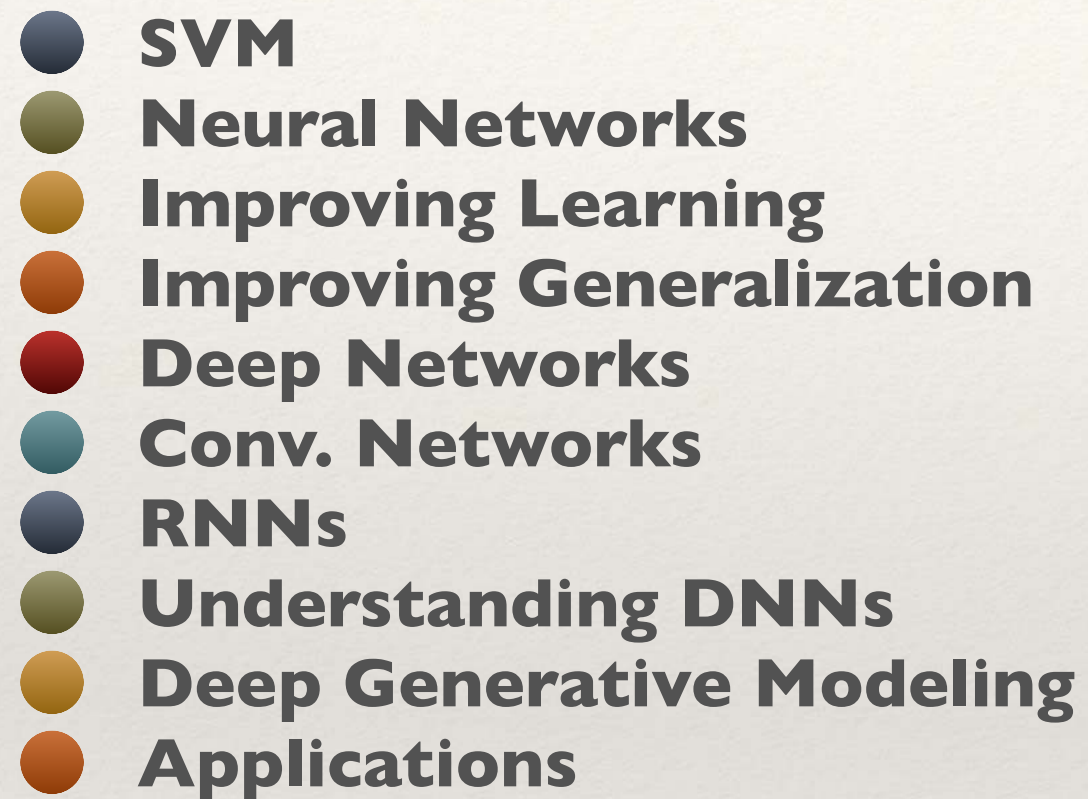


Generative Modeling and Dimensionality Reduction

- **Feature Processing**
- **PCA/LDA**
- **Gaussian and GMM**
- **NMF**
- **Linear and Logistic Regression**
- **kernel methods**



Discriminative Modeling



When we started ...

Dates of Various Rituals

- ❖ 5 Assignments spread over 3 months (roughly one assignment every two weeks).
- ❖ September 1st week - project topic announcements.
- ❖ September 3rd week - 1st Midterm
- ❖ September 4th week - project topic and team finalization and proposal submission. [1 and 2 person teams].
- ❖ October 1st week - Project Proposal
- ❖ October 3rd week - 2nd MidTerm
- ❖ November 1st week - Project MidTerm Presentations.
- ❖ **December 1st week - Final Exams**
- ❖ **December 2nd week - Project Final Presentations.**

Content Delivery

